HIGH-RESOLUTION IMAGE RECONSTRUCTION USING DEEP NEURAL NETWORKS

1. Introduction

Image reconstruction is the process of recovering an image from incomplete, compressed, or limited information. The main objective of this project is to investigate how an image can be reconstructed from a reduced number of measurements while maintaining its important visual information. Instead of directly processing all the pixels of an image, a measurement matrix is used to generate a lower-dimensional representation.

The project primarily investigates a Generative Adversarial Network (GAN)-based reconstruction approach. Conventional sparse-recovery techniques, including Least Absolute Shrinkage and Selection Operator (LASSO) and Sparse Bayesian Learning (SBL), are considered as reference methods for evaluating reconstruction performance.

2. Problem Statement and Objectives

High-dimensional images contain many pixel values, increasing the computational and storage requirements of image processing. The project investigates whether an original image can be reconstructed accurately from fewer measurements using a GAN-based model. Another objective is to study how reconstruction quality changes when the number of measurements is reduced.

The measurement process is represented mathematically as:

$$\mathbf{y = Ax}$$

Here, x denotes the original image vector, A denotes the measurement matrix, and y denotes the compressed measurement vector. The reconstruction model uses the measurements to estimate the original image.

The main objectives are to implement GAN-based image reconstruction, evaluate the effect of measurement reduction, compare the results with LASSO and SBL, and investigate the suitability of the approach for additional image datasets.

3. Literature Review and Initial Learning

The initial literature study included *Sparse Signal Recovery Using Deep Neural Networks* by Rubin Jose Peter and Chandra R. Murthy. This paper provided a reference for understanding signal recovery from sparse measurements. Additional studies considered GAN-based compressed sensing, single-pixel imaging, MRI reconstruction from undersampled k-space data, and ultra-sparse-view lung CT reconstruction.

During the initial learning stage, the fundamental concepts of Deep Neural Networks were studied, including input, hidden and output layers, neurons, weights, activation functions, forward propagation, loss functions, and model training. This learning established the foundation for understanding how a neural network learns the relationship between input measurements and a desired reconstructed image.

The subsequent learning stage focused on GAN architecture and the roles of its two main components: the generator and discriminator.

4. Methodology and Experimental Setup

The initial experiments use the MNIST handwritten-digit dataset. Each image has a resolution of 28 × 28 pixels, giving 784 pixel values. The images are normalized to the range 0–1 and represented as vectors.

A measurement matrix compresses the original image into a smaller measurement vector. The generator receives these measurements and produces a reconstructed image. The discriminator distinguishes original dataset images from generated images. During training, the generator learns to reduce reconstruction error while improving the realism of its output through adversarial training.

The implementation uses Python, NumPy, TensorFlow/Keras, and Matplotlib for numerical operations, model construction, training, and visualization. The Adam optimizer is used to update the trainable network parameters.

5. Experimental Results and Performance Evaluation

Initial experiments were conducted using different measurement dimensions. The presentation reports the following results:

Experimental setting Reported Normalized MSE

N = 784, M = 100 0.12024844

N = 784, M = 50 0.30164862

These results show that the reported normalized reconstruction error increases when the measurement dimension decreases from 100 to 50. This indicates that reconstruction becomes more challenging when less measurement information is available.

Reconstruction performance is evaluated using Mean Squared Error (MSE), Normalized Mean Squared Error (NMSE), and NMSE in decibels. Visual comparisons between original and reconstructed images are also used to examine the preservation of image structure.

LASSO and SBL are included as conventional reference methods. LASSO uses L1 regularization to encourage sparse solutions, whereas SBL uses a Bayesian framework for sparse signal recovery. Their purpose is to provide reference points for evaluating the GAN-based approach.

6. Future Work and Conclusion

The next stage involves extending the experiments beyond MNIST. CIFAR-10 is being considered as an additional dataset. Its RGB images can be converted to grayscale using the luminance method:

Y = 0.299R + 0.587G + 0.114B

This converts each 32 × 32 RGB image into a single-channel grayscale image containing 1,024 pixel values. The images can then be normalized, converted into measurement vectors, and used in the reconstruction framework. This extension is planned work and is not included among the completed MNIST results reported above.

Future work includes testing the GAN, LASSO, and SBL approaches on additional datasets, evaluating different measurement dimensions, comparing reconstruction quality quantitatively and visually, and investigating possible model improvements.

In conclusion, the project studies image reconstruction from low-dimensional measurements using a GAN-based approach, with LASSO and SBL serving as conventional comparison methods. The initial MNIST experiments provide a basis for investigating reconstruction quality under different measurement conditions and extending the framework to other image datasets.

References

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